

PEDIGREES

Pedigrees or family trees show how a particular characteristic is passed down through several generations of a family.

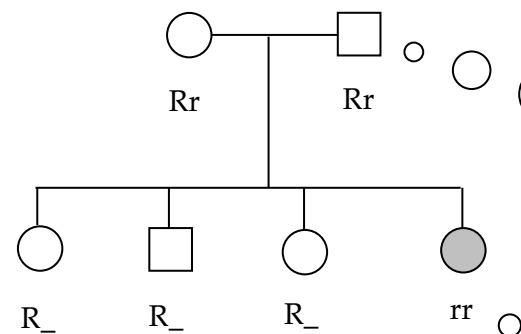
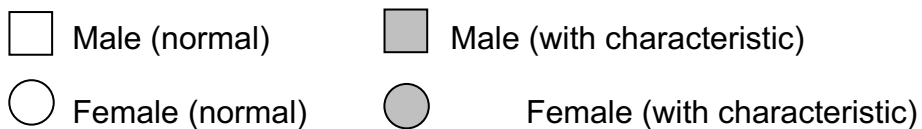
In determining whether a particular characteristic is **dominant** or **recessive**, look at the **parents and their offspring at any part** of the pedigree.

If both parents **DON'T SHOW** the characteristic, and at least one offspring **DOES**, then the characteristic is **RECESSIVE**.

If both parents **DO SHOW** the characteristic, and at least one offspring **DOESN'T**, then the characteristic is **DOMINANT**.

Example

Consider the incidence of **left-handedness** in a family.



Both parents don't show the characteristic but one child does.

∴ Left-handedness is **recessive**.

Second allele is unknown until these offspring have their own children.

One child is left-handed.

∴ Genotype is **rr**.

Both parents must be **hybrid** - each one contributes an **r** allele.

Hint

Always **start at the bottom of a pedigree** and work your way upward when determining the genotypes of all the individuals.

Example

The following pedigree shows the incidence of attached earlobes in a family. Analyse the diagram to show:

- (a) whether the characteristic is dominant or recessive.
- (b) the genotype for every individual.

